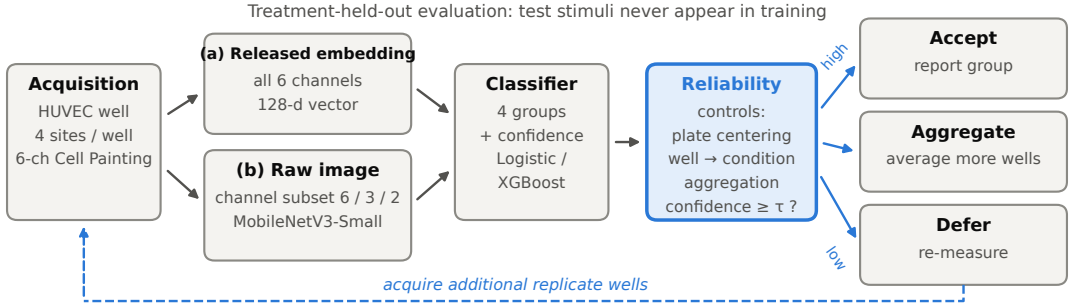
Treatment-held-out evaluation: test stimuli never appear in training
Acquisition
HUVEC well
4 sites / well
6-ch Cell Painting
(a) Released embedding
all 6 channels
128-d vector
(b) Raw image
channel subset 6 / 3 / 2
MobileNetV3-Small
Classifier
4 groups
+ confidence
Logistic /
XGBoost
Reliability
controls:
plate centering
well → condition
aggregation
confidence ≥ τ ?
high
low
Accept
report group
Aggregate
average more wells
Defer
re-measure
acquire additional replicate wells